Pre-analysis plan

RT

This pre-analysis plan describes how I will empirically test the four hypotheses about the effects of having daughters using data from the German Socio-Economic Panel (SOEP).

Hypothesis 1

H1. *Having daughters leads to stronger support for gender equality (as measured by attitudes)*. That is, denoting *X* as a measure of support for gender equality, (*X* for respondents with daughters) > (*X* for respondents without daughters).

- **Variables**
  - Dependent variable: Fam: Man/Women Equal in Household/Family and Job (lb1120), recoded to range between 0 (completely disagree) and 1 (agree completely) [$support_gender_equality$].
  - Independent variable: Dummy variable for whether participant has at least one daughter (1) or not (0) [$d_daughter$].
  - Control variables: Number of children [n_kids], gender [d_sex], age [age], and survey year [$syear$] fixed effects [δ_t].

- **Inclusion/exclusion criteria**
  - Include only participants [i] with at least one child in the survey year [t].
  - Exclude observations with missing information on any of the variables included in the regression.
  - Include all years between 2000 and 2020 with available data for the relevant variables.

- **Exact analysis**
  - $support_gender_equality_{it} = \beta_0 + \beta_1 d_daughter_{it} + \beta_2 n_kids_{it} + \beta_3 d_sex_{it} + \beta_4 age_{it} + \delta_t + \varepsilon_{it}$
  - Robust standard errors clustered at the household [hid] level.

- **Interpretation of results**
  - Hypothesis supported if β_1 positive and statistically significant (p-value < 0.05).

Hypothesis 2

H2. *Having daughters leads to more left-leaning political preference.* That is, denoting *Y* as a measure of left-leaning political preferences, (*Y* for respondents with daughters) > (*Y* for respondents without daughters).

- **Variables**
  - Dependent variable: Political attitude (plh0004), recoded to range between 0 (entirely right) and 1 (entirely left) [$left_political_preferences$].
  - Independent variable: Dummy variable for whether participant has at least one daughter (1) or not (0) [$d_daughter$].
  - Control variables: Number of children [n_kids], gender [d_sex], age [age], and survey year [$syear$] fixed effects [δ_t].

- **Inclusion/exclusion criteria**
  - Include only participants [i] with at least one child in the survey year [t].
  - Exclude observations with missing information on any of the variables included in the regression.
  - Include all years between 2000 and 2020 with available data for the relevant variables.

- **Exact analysis**
  - $left_political_preferences_{it} = \beta_0 + \beta_1 d_daughter_{it} + \beta_2 n_kids_{it} + \beta_3 d_sex_{it} + \beta_4 age_{it} + \delta_t + \varepsilon_{it}$
  - Robust standard errors clustered at the household [hid] level.

- **Interpretation of results**
  - Hypothesis supported if β_1 positive and statistically significant (p-value < 0.05).

Hypothesis 3

H3. *Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).* That is, denoting *Z* as a measure of gender equality in the household, (*Z* for respondents with daughters) > (*Z* for respondents without daughters).

- **Variables**
  - Dependent variable: Absolute value of difference in Actual Work Time Per Week (pgtatzeit) between men and women in the household, recoded to range between 0 (gender unequal) and 1 (gender equal) [$gender_equality_household$].
  - Independent variable: Dummy variable for whether participant has at least one daughter (1) or not (0) [$d_daughter$].
  - Control variables: Number of children [n_kids], gender [d_sex], age [age], and survey year [$syear$] fixed effects [δ_t].

- **Inclusion/exclusion criteria**
  - Include only participants [i] with at least one child in the survey year [t].
  - Include only households with positive work time for both men and women.
  - Exclude observations with missing information on any of the variables included in the regression.
  - Include all years between 2000 and 2020 with available data for the relevant variables.

- **Exact analysis**
  - $gender_equality_household_{it} = \beta_0 + \beta_1 d_daughter_{it} + \beta_2 n_kids_{it} + \beta_3 d_sex_{it} + \beta_4 age_{it} + \delta_t + \varepsilon_{it}$
  - Robust standard errors clustered at the household [hid] level.

- **Interpretation of results**
  - Hypothesis supported if β_1 positive and statistically significant (p-value < 0.05).

Hypothesis 4

H3. *Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).* That is, denoting *X* as a measure of support for gender equality, (*X* for male respondents with daughters – *X* for male respondents without daughters) > (*X* for female respondents with daughters – *X* for female respondents without daughters).

- **Variables**
  - Dependent variable: Fam: Man/Women Equal in Household/Family and Job (lb1120), recoded to range between 0 (completely disagree) and 1 (agree completely) [$support_gender_equality$].
  - Independent variable: Dummy variable for whether participant has at least one daughter (1) or not (0) [$d_daughter$] interacted with dummy variable for male (1) or female (0) gender [d_sex].
  - Control variables: Daughter [$d_daughter$], gender [d_sex], number of children [n_kids], age [age], and survey year [$syear$] fixed effects [δ_t].

- **Inclusion/exclusion criteria**
  - Include only participants [i] with at least one child in the survey year [t].
  - Exclude observations with missing information on any of the variables included in the regression.
  - Include all years between 2000 and 2020 with available data for the relevant variables.

- **Exact analysis**
  - $support_gender_equality_{it} = \beta_0 + \beta_1 d_daughter_{it} + \beta_2 d_sex_{it} + \beta_3 (d_daughter_{it} \times d_sex_{it}) + \beta_4 n_kids_{it} + \beta_5 age_{it} + \delta_t + \varepsilon_{it}$
  - Robust standard errors clustered at the household [hid] level.

- **Interpretation of results**
  - Hypothesis supported if β_3 positive and statistically significant (p-value < 0.05).